

Ziyi Sun

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EDUCATION

- **Ocean University of China (985 / Double First-Class)** Sep. 2023 – Jun. 2027 (Expected)
B.Eng. in Mechanical Design, Manufacturing and Automation Qingdao, China
 - GPA: **3.7/4.0** Rank: **1/56** Average: 89.2
 - Selected courses: Advanced Mathematics (97), Linear Algebra (93.5), Probability & Mathematical Statistics (93), Complex Variables & Integral Transforms (94.5), Numerical Methods (96), Mechanical Vibrations (95), Modern Mechanical Design Theory and Methods (100), Underwater Robotics (97.2)

RESEARCH EXPERIENCE

- **Roboparty Lab** Jul. 2026 – Oct. 2026
Frontier Research and Open Source Intern (Mentor: Jagger) Beijing, China
 - Contribute to frontier research and open-source development for embodied intelligence and robot learning systems.

PROJECTS

- **Variable-Length Multimodal Biomimetic Robotic Fish for Deep-Sea Inspection** Dec. 2025 – Dec. 2026
Role: Project Leader | Tools: SolidWorks, multimodal vision & sensing OUC SRDP
 - Designed a biomimetic robotic fish with continuously adjustable body length (1.2–1.5 m) for deep-sea cage cruise and narrow-space inspection.
 - Fused school-fish visual tracking with water-quality sensing for joint monitoring and early warning of aquaculture risks.
- **Cross-Domain Wire-Driven Biomimetic Eel Robot** Dec. 2024 – Nov. 2025
Role: Project Leader | Tools: vision tracking, Gaussian Splatting, damage detection Provincial Innovation Program
 - Developed an eel-like robot with cruising and active depth control for floating offshore wind dynamic-cable inspection.
 - Integrated visual tracking, 3D reconstruction, and damage detection for cable modeling, defect localization, and risk warning.
- **Modular Adaptive Deep-Sea Resource Collection and Locomotion Platform** Dec. 2024 – Nov. 2025
Role: Core Member | Tools: Fluent, mechanical design Research Project
 - Designed a modular adaptive deep-sea mining vehicle for complex seabed terrain, including Fluent-based plume analysis during locomotion/mining and plume-suppression structures.
 - Responsible for analysis and design of the compliant collection structure; combined with an active four-track chassis to improve geological adaptability.
- **High-Stability Duck-Type Wave Energy Converter with Pendulum PTO** Dec. 2024 – Nov. 2025
Role: Core Member | Tools: AQWA, genetic algorithm, SolidWorks National Innovation Program
 - Improved capture efficiency and output stability via hydrodynamics modeling, shape optimization, and a combined PTO with nonlinear adaptive regulation.

PUBLICATIONS & PATENTS

C=CONFERENCE, S=IN SUBMISSION, P=PATENT, SW=SOFTWARE

[C.1] A Two-Stage Learning Framework for Autonomous Obstacle Avoidance of an Eel-Like Robot.

Ziyi Sun, Luwen Li, Changhong He, Shuo Jiang, Zhuoyan Wang, Haoyuan Cheng*.

IEEE International Conference on Real-time Computing and Robotics (IEEE RCAR), 2026. **Best Paper Finalist.**

- Proposed a two-stage framework combining multimodal perception, expert demonstration, PPO, AHTA-CPG, and domain randomization.
- Achieved 70%–90% success over 60 real water-tank trials.

[C.2] Design of Biomimetic Robotic Eel with Omnidirectional Wire-Driven Body and Controlled Central Pattern Generator.

Dingyang Yu, Yuanrong Zhong, Qi Chen, Siming Ke, Qianbei Luo, Ziyi Sun, Xunqi Zhao.

IEEE International Conference on Computer, Control and Robotics (ICCCR), 2025.

◦ Presented the mechanical design and control framework of a biomimetic robotic eel with an omnidirectional quadruple wire-driven body and CPG-based locomotion control.

[S.1] **CHEROE: Every Humanoid Skill as a Traject.**
Ziyi Sun, Jingwen Chen, Yuxi Wang, Xiuze Xia, Long Cheng, Zhaoxiang Zhang, Junyu Dong.
Under review at AAAI 2027 (CCF-A).

◦ Unified SkillMotion representation and Tracker interface with boundary compatibility assessment, trajectory fusion, and bridging-action retrieval for reusable multi-source humanoid skills.

[S.2] **A Streaming Evaluation Framework for Deep-Sea Mining: Comparing Centralized and Distributed Nodule Collection.**
Ziyi Sun, Changhong He, et al.
Under review at *Journal of Marine Science and Engineering (JMSE)*.

◦ Developed a unified simulation and benchmarking pipeline for scalable robot learning, integrating CAD-based robot models and reproducible comparisons of centralized vs. distributed multi-AUV nodule collection.

[P.1] **A High-Stability PTO System for Duck-Type Wave Energy Converters.**
Zhi Han, *Ziyi Sun*, Haoran Chen, Hailong Zhao, Wenchi Wei, Xueyao Wang, Hongda Shi.
Invention Patent (First Student Inventor).

[SW.1] **Underwater Polarized Light Field Prediction Software for Biomimetic Navigation V1.0.**
Haoyuan Cheng, Chuanyun Su, Zhichao Yan, Shuning Wang, *Ziyi Sun*.
Software Copyright (No. 2024SR0140235).

SKILLS

- **Robot Platform:** Unitree G1
- **Programming & ML:** Python, C/C++, MATLAB, PyTorch, Isaac Gym/Lab, MuJoCo
- **Mechanical Design:** SolidWorks, AutoCAD
- **Languages:** Chinese (Native); English (CET-4: 591, CET-6: 482)

HONORS AND AWARDS

• Shandong Provincial Government Scholarship	2025
• First-Class Comprehensive Scholarship, Ocean University of China	2024 & 2025
• Outstanding Student / Outstanding Student Cadre, Ocean University of China	2024 & 2025
• National First Prize, National 3D Digital Innovation Design Competition	2025 & 2026
• National Second Prize, Contemporary Undergraduate Mathematical Contest in Modeling (CUMCM)	2025
• Honorable Mention, Mathematical Contest in Modeling (MCM)	2025
• National Third Prize, 18th National College Student Energy Conservation Competition	2024
• National Gold Award, 3rd “Chuangyi Cup” National Undergraduate Extracurricular Academic Contest	2024
• National Second Prize, 14th Marine Vehicle Design and Production Competition	2025
• National First Prize, CRRC Cup National Undergraduate Renewable Energy Competition	2026
• Science and Technology Innovation Pioneer Team, College of Engineering, OUC	2024

LEADERSHIP EXPERIENCE

• Ocean University of China	2023 – Present
Class Representative / Class Monitor	Qingdao, China
• College of Engineering Debate Team, OUC	2023 – Present
Team Leader	Qingdao, China
• Student Association for the Study of Socialism with Chinese Characteristics, OUC	2023 – Present
Director of Event Organization	Qingdao, China

VOLUNTEER EXPERIENCE

• Volunteer, Qingdao Marathon	2023
• Member, National College Student Volunteer Outreach Team	2024
• Outstanding Individual in Summer Social Practice, OUC	2024